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Circulation

ORIGINAL RESEARCH ARTICLE

Transcatheter Aortic Valve Replacement With Balloon- Versus Self-Expandable Bioprostheses for the Treatment of Bicuspid Aortic Valve Stenosis

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BACKGROUND: Differences between balloon- and self-expandable transcatheter heart valves (BE-THVs and SE-THVs, respectively) may influence the outcomes of transcatheter aortic valve replacement for bicuspid aortic valve (BAV) stenosis.

METHODS: Consecutive patients undergoing transcatheter aortic valve replacement with BE-THV or SE-THV for computed tomography–diagnosed bicuspid aortic valve stenosis at 29 centers were included. The primary outcome was death or stroke. After propensity score matching in 10 data sets generated by multiple imputation, outcomes from transcatheter aortic valve replacement to 3-year follow-up were computed by multivariable binomial logistic mixed-effects models, multivariable linear mixed-effects models, or multivariable frailty models accounting for center-related influences and residual confounding effects (doubly robust adjustment). The results were replicated by inverse probability of treatment weighting and multivariable adjustment.

RESULTS: A total of 1443 consecutive patients with bicuspid aortic valve stenosis undergoing BE-THV (n=860) or SE-THV (n=583) implantation were included. In-hospital and 30-day death or stroke did not significantly differ between BE-THV and SE-THV groups (5.1% versus 6.1%; hazard ratio after propensity score matching, 1.02 [95% CI, 0.51–2.02]). BE-THV implantation was associated with higher annulus rupture and mean transvalvular gradient compared with SE-THV implantation. In contrast, SE-THV implantation was associated with higher additional valve implantation and paravalvular regurgitation compared with BE-THV implantation. The results were consistent across the statistical methods used and between early- and new-generation THVs. At 30 days, pacemaker implantation was lower in the BE-THV group compared with the SE-THV group (11.9% versus 18.6%; hazard ratio after propensity score matching, 0.58 [95% CI, 0.36–0.93]). This result did not depend on the statistical method used. At 3 years, consistent with the 1- and 2-year analyses, death or stroke was not significantly different between the BE-THV and SE-THV groups (23.7% versus 26.2%; hazard ratio after

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propensity score matching, 0.99 [95% CI, 0.65–1.51]). Death or stroke across major clinical, anatomical, functional, and procedural conditions was consistent with the main analysis. After inverse probability of treatment weighting and multivariable adjustment, these conclusions remained unchanged.

CONCLUSIONS: In patients undergoing transcatheter aortic valve replacement for bicuspid aortic valve stenosis, death or stroke does not significantly differ between those receiving a BE-THV and those receiving an SE-THV over a follow-up of 3 years. BE-THV is associated with higher transvalvular mean gradient and more frequent annulus rupture, whereas SE-THV is associated with more frequent moderate to severe aortic regurgitation, additional THV implantation, and permanent pacemaker implantation.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: aortic valve stenosis ■ bicuspid aortic valve disease ■ transcatheter aortic valve replacement

Editorial, see p 658

Clinical Perspective

What Is New?

- This observational study shows that in a population of patients with computed tomography–diagnosed bicuspid aortic valve stenosis who underwent transcatheter aortic valve replacement, balloon-expandable transcatheter heart valve (BE-THV) implantation is associated with incidences of death or stroke comparable to self-expandable transcatheter heart valve (SE-THV) implantation over a follow-up of 3 years.
- Patients who received a BE-THV experienced annulus rupture and postprocedural aortic mean gradient ≥20 mm Hg more frequently than those who received an SE-THV.
- Patients who received an SE-THV required additional transcatheter heart valve implantation and experienced postprocedural moderate to severe aortic regurgitation more frequently than those who received a BE-THV.
- Patients who received a BE-THV required pacemaker implantation less frequently than those who received an SE-THV.

What Are the Clinical Implications?

- The type of THV deployment mechanism and inherent technical THV features do not seem to influence major cardiovascular outcomes over a follow-up of 3 years.
- Nevertheless, the use of BE-THV and SE-THV seems to be associated with different acute complications, particularly annulus rupture and additional transcatheter heart valve implantation, implying some considerations about the case-specific anatomy in the selection of the type of transcatheter heart valve to implant.
- The long-term impact of moderate to severe aortic regurgitation after SE-THV implantation warrants further analysis.

Nonstandard Abbreviations and Acronyms

BAV	bicuspid aortic valve
BE-THV	balloon-expandable transcatheter heart valve
HR	hazard ratio
IPTW	inverse probability of treatment weighting
MVA	multivariable adjustment
OR	odds ratio
PSM	propensity score matching
SE-THV	self-expandable transcatheter heart valve
STS/ACC TVT	Society of Thoracic Surgeons/American College of Cardiology Transcatheter Valve Therapies
STS-PROM	Society of Thoracic Surgeons–predicted risk of mortality
TAVR	transcatheter aortic valve replacement
THV	transcatheter heart valve

Bicuspid aortic valve (BAV) is a common congenital heart disease, affecting ≈0.5% to 2.0% of the population.[1–5] This condition has been associated with a higher incidence of cardiovascular complications throughout the life span because of the marked propensity to develop valve stenosis, insufficiency, endocarditis, and ascending aorta aneurysm and dissection.[1–5] Among BAV disease patterns, the most common phenotype in adults is stenosis.[6–8] Significant proportions of these patients eventually require aortic valve replacement, generally 10 years earlier than patients with tricuspid aortic valve, with the highest incidence occurring in the sixth decade.[6–8] However, the prevalence of elderly patients with BAV stenosis is not negligible, and before

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the transcatheter aortic valve replacement (TAVR) era, almost 25% of octogenarians referred to surgery for isolated valve replacement had BAV stenosis.[7,8]

Almost all randomized trials comparing TAVR with surgical aortic valve replacement, including PARTNER 3 (Safety and Effectiveness of the SAPIEN 3 Transcatheter Heart Valve in Low Risk Patients With Aortic Stenosis) and Evolut Low Risk, have systematically excluded patients with BAV because of anatomical and technical concerns.[9–16] For these reasons, current American Heart Association/American College of Cardiology and European Society of Cardiology guidelines preferentially recommend surgical aortic valve replacement for most patients with symptomatic BAV stenosis.[9,10] However, the approval of TAVR for low-risk patients and the real-world practice expansion into younger and less comorbid populations have substantially increased the number of TAVR interventions for BAV stenosis.[15–17] The growing interest in this subset is supported by recent data showing that yearly proportions of TAVR for BAV stenosis in high-volume centers exceed 10%.[17,18]

Most of the available studies on TAVR for BAV stenosis consist of small cohorts with short-term follow-up.[19,20] A few more recent, larger studies exploring the performance of TAVR between BAV and tricuspid aortic valve reported comparable incidences of death or stroke at 30 days and 1 year between settings.[18,21,22] However, these studies were conducted to independently assess the outcomes after TAVR in a population of patients receiving either a balloon-expandable (BE-) transcatheter heart valve (THV) or a self-expandable (SE-THV), thereby precluding head-to-head comparisons of THV technologies.[18,21,22] In the only study from the STS/ACC TVT registry (Society of Thoracic Surgeons/American College of Cardiology Transcatheter Valve Therapies) encompassing both types of THV, BE-THV and SE-THV groups were not compared, and available assessments of patients with TAVR within the BAV setting essentially refer to the comparison of early- and new-generation THVs.[23]

In the setting of BAV stenosis, the deployment method and inherent technical features of BE-THV and SE-THV have been associated with potentially relevant differences in procedural and midterm clinical outcomes.[3,24,25] Yet, these considerations essentially remain empirical because of a paucity of data.[3,24–26] Against this background, we conducted a large multicenter observational study to compare early and late outcomes between patients with BAV stenosis who underwent BE-THV implantation and those who underwent SE-THV implantation.

METHODS

Consecutive patients with aortic valve stenosis and computed tomography–diagnosed BAV who underwent TAVR with BE-THV or SE-THV at 29 centers were screened. Participating centers were characterized by the presence of a long-time established interventional program and a dedicated institutional database prospectively capturing relevant clinical, anatomical, functional, and procedural data independently from the present study. The inclusion criteria were the following: computed tomography–defined BAV, severe aortic stenosis, and BE-THV or SE-THV implantation. No clinical, anatomical, functional, or procedural restrictions were imposed, but several subgroup and sensitivity analyses were planned. Informed consent was obtained from all patients before the procedure; individual anonymized data were collected by dedicated electronic spreadsheets; and local ethics committees approved data sharing. The data underlying this article will be shared on reasonable request to the corresponding author. The study was reported according to the Strengthening the Reporting of Observational Studies in Epidemiology recommendations (Table S1).[27]

Definitions

The definitions used in the study (ie, BAV type and subtype, SE-THV, BE-THV, THV generation, functional grading, etc) are described extensively in the Supplemental Material.

Outcomes

Outcomes were assessed from TAVR to 3-year follow-up. Outcome definitions were based on the VARC-3 (Valve Academic Research Consortium) recommendations.[28] The primary outcome across in-hospital, 30-day, 1-year, 2-year, and 3-year analyses was a composite of death or stroke. Secondary outcomes across follow-up times included death, cardiac death, stroke, vascular complications, bleeding, acute kidney injury, repeat aortic valve replacement, valve thrombosis, endocarditis, pacemaker implantation, and valve-related rehospitalization.[28]

Statistical Analysis

Statistical analysis is more extensively described in the Supplemental Material. Categorical variables were summarized as counts and percentages. Continuous variables were summarized as mean and standard deviation or median and interquartile range according to the distribution as assessed by the Shapiro-Wilk test. Baseline and procedural variables balance between BE-THV and SE-THV groups was individually assessed primarily by standardized mean differences as recommended for studies with adjustment based on the propensity score.[29] By convention, a standardized mean difference within the range of ±0.100 indicates a good balance.[29] A formal assessment of between-group balance of continuous variables by the Welch *t* test or Wilcoxon-Mann-Whitney *U* test, depending on their distribution, and categorical variables by Pearson χ^2 or Fisher exact test was supplemented.

In the unadjusted analysis, in-hospital categorical outcomes computed by binomial logistic regression were presented by odds ratios (ORs) with 95% CIs, and continuous outcomes computed by linear regression were presented by coefficients with 95% CIs.[30] Unadjusted incidences of 30-day, 1-year, 2-year, and 3-year outcomes were computed by the Kaplan-Meier method and compared by the log-rank test.[31] Unadjusted hazard ratios (HRs) with 95% CIs were computed by Cox proportional hazards regression.[31] The proportional hazards assumption was assessed by the Grambsch-Therneau test and

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Schoenfeld scaled residuals.[31] In the case of significant violations of the proportional hazards assumption, the use of time-varying models was planned.[31]

Multivariate imputation by chained equations was used to impute missing data in the database through an iterative series (10 iterations) of predictive models based on the observed values to generate a total of 10 imputed data sets.[32–34] More details are provided in the Supplemental Material.

Three adjusted analyses were independently performed: propensity score matching (PSM), inverse probability of treatment weighting (IPTW), and multivariable adjustment (MVA). PSM involves identifying a subset of subjects from the unexposed sample whose covariate distribution closely matches that of subjects from the exposed sample based on the propensity score, a 1-dimensional summary measure of the individual probability of treatment assignment conditional on observed baseline characteristics.[35–40] Propensity score was computed across the imputed data sets with a generalized linear model, binomial family, and logit link, including measured baseline characteristics as covariates and THV type (BE-THV or SE-THV) as the dependent variable.[35–40] The following covariates were included in the model: age (continuous), sex, body mass index (continuous), diabetes, hypertension, coronary artery disease, previous myocardial infarction, previous percutaneous coronary intervention, previous coronary artery bypass grafting, previous pacemaker implantation (except for the outcome of pacemaker implantation), previous stroke, peripheral artery disease, chronic obstructive pulmonary disease, estimated glomerular filtration rate (continuous), New York Heart Association class, left ventricular ejection fraction (continuous), aortic valve mean gradient by echocardiography (continuous), aortic valve area by echocardiography (continuous), peak systolic velocity by echocardiography (continuous), aortic regurgitation grade by echocardiography, mitral regurgitation grade by echocardiography, tricuspid regurgitation grade by echocardiography, BAV type by computed tomography, aortic annulus area by computed tomography by computed tomography (continuous), aortic valve leaflets calcification by computed tomography, aortic valve annulus calcification by computed tomography, aortic valve left ventricular outflow calcification by computed tomography, ascending aorta diameter by computed tomography (continuous), procedure year, procedural timing, and Society of Thoracic Surgeons–predicted risk of mortality (STS-PROM; continuous). A 1:1 nearest neighbor matching without replacement, using a caliper width of 0.1 and targeting the average treatment effect on the treated, was used across the imputed data sets.[35–40] The within-data set method was used for combining multiple imputation with PSM because it showed superior performance compared with the across–data set method.[41] The balance of covariates between the unexposed and exposed samples before and after PSM in each of the 10 data sets generated by multiple imputation was assessed by the standardized mean difference, variance ratio, and Kolmogorov-Smirnov statistics and displayed by Love plot, histograms, densities, and empirical cumulative distribution functions.[29,40] In-hospital categorical and continuous outcomes were computed by binomial logistic mixed-effects models and linear mixed-effects models, respectively.[42] Thirty-day, 1-year, 2-year, and 3-year outcomes were computed by semiparametric penalized partial likelihood approach–based shared log-normal frailty models with a random intercept to account for differences in the baseline hazard across centers and a random slope to account for between-center differences in treatment effects, including a higher use of a type of THV over the other.[31,43–47] All these models used to estimate the outcomes were multivariable. Specifically, according to the "doubly robust" adjustment, in addition to the type of THV (BE-THV versus SE-THV), they included baseline clinical, anatomical, and functional covariates used to estimate the propensity score and potential procedural confounders (ie, access site, anesthesia, predilation, THV generation, postdilation, and access closure technique).[48] The outcome estimates between treatment groups across the imputed data sets were combined according to the Rubin rules.[32–34]

IPTW and MVA, the other 2 adjustment methods used to evaluate the consistency of results, are described in the Supplemental Material.[38,39,49] IPTW and MVA were applied across the imputed data sets.[38,39,49] As described for PSM, outcomes after IPTW were estimated by consistent mixed-effects models, including baseline clinical, anatomical, functional, and procedural confounders according to the "doubly robust" adjustment.[31,42–48] MVA was performed by consistent mixed-effects models, including the same baseline variables used for PSM and IPTW.[31,42–48] The outcome estimates between treatment groups across the imputed data sets were combined by the Rubin rules.[32–34]

Adjusted survival curves and survival estimates after PSM, IPTW, and MVA within each data set generated by multiple imputation were computed in a multistage process. More details are provided in the Supplemental Material .

Prespecified subgroup analyses were conducted to explore the influence of major subsets on the primary outcome of death or stroke. More details are provided in the Supplemental Material. In additional subgroup analyses, the 2 key postprocedural outcomes of aortic valve mean gradient and aortic valve regurgitation and the 30-day outcome of permanent pacemaker implantation were assessed according to the implantation of early- or new-generation THVs.

A sensitivity analysis focusing only on the most widely implanted types of BE-THV and SE-THV was performed, also prompted by the recent approval in the United States and Europe for the treatment of selected patients with BAV stenosis with these THVs. Additional sensitivity analyses focusing only on patients with assessable computed tomography scan and patients with BAV type 1 or 2 characterized by moderate to severe raphe(s) and leaflets calcification were performed.

Clinical, anatomical, functional, and procedural variables with potential association with procedural annulus rupture, post-TAVR moderate to severe aortic mean gradient, post-TAVR moderate to severe aortic regurgitation, and in-hospital stroke were investigated by multivariable models. Aortic mean gradient and regurgitation at the 1-year (±3 months) follow-up assessed by transthoracic echocardiography were used to describe functional variations between groups over time.

Finally, a landmark analysis to assess between-group trends in major outcomes within the period from the 1- to 3-year follow-up was performed to complement the main analysis.

All *P* values were 2 sided, and 0.05 was the threshold for declaring statistical significance. The analyses were conducted with R statistical software (version 4.3.2).

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RESULTS

The study included 1443 consecutive patients treated with BE-THV (n=860, 59.6%) or SE-THV (n=583; 40.4%) from May 2007 to December 2021. Baseline clinical, anatomical, functional, and procedural characteristics are reported in Table 1. Overall, the median age was 78.7 years; 39.4% of the patients were female; BAV type 1 was the predominant morphology (88.4%); median STS-PROM was 3.2%; and 91.2% of the TAVR procedures were performed from 2014 to 2021. Information on THVs and data completeness is reported in Tables S2 through S5. After PSM, there were overall no significant between-group differences as denoted by standardized mean differences within the range of ±0.100 (Table 1; Figure 1; Tables S6 and S7; Figures S1 through S4). Consistently, after IPTW, there were overall no significant between-group differences (Tables S8 through S10; Figures S5 through S8).

Short-Term Outcomes

In-hospital outcomes are reported in Table 2 and Table S11. Death or stroke was not significantly different between patients who received a BE-THV and those who received an SE-THV (unadjusted rates: 4.0% versus 5.7%; OR, 0.69 [95% CI, 0.42–1.13]; OR_{PSM}, 0.93 [95% CI, 0.45–1.93]; OR_{IPTW}, 0.79 [95% CI, 0.45–1.40]; OR_{MVA}, 0.88 [95% CI, 0.47–1.65]), as well the individual components of the composite outcome (Table 2; Table S11). BE-THV implantation was more frequently associated with annulus rupture compared with SE-THV implantation (average PSM-based rates, 0.5 % versus 1.4%), whereas patients who received an SE-THV required additional THV implantation more frequently than those who received a BE-THV (Table 2; Table S11). Fewer than half of patients who experienced annulus rupture underwent intraoperative or early postprocedural surgery with an unfavorable outcome, and the remaining patients were managed conservatively because of self-limiting extravasation. BE-THV implantation was associated with a higher risk of postprocedural mean gradient >20 mm Hg compared with SE-THV implantation (averaged PSM-based rates, 6.0% versus 2.3%; $P<0.001$), whereas BE-THV implantation was associated with a lower occurrence of moderate to severe postprocedural aortic valve regurgitation compared with SE-THV implantation (averaged PSM-based rates, 3.6% versus 13.2%; $P<0.001$; Table 2; Figure 2; Table S11; Figure S9). The 2 outcomes were also assessed according to the generation of THV, and no significant treatment-by-subgroup interaction was detected, regardless of the adjustment method used (Table S12). In-hospital pacemaker implantation was significantly less frequent in patients who received a BE-THV compared with those who received an SE-THV (Table 2; Table S11). There were no significant adjusted differences in the other in-hospital outcomes (Table 2; Table S11).

At 30 days, death or stroke was not significantly different between the BE-THV and SE-THV groups (unadjusted rates, 5.1% versus 6.1%; HR, 0.83 [95% CI, 0.53–1.29]; HR_{PSM}, 1.07 [95% CI, 0.55–2.07]; HR_{IPTW}, 1.05 [95% CI, 0.63–1.73]; HR_{MVA}, 1.00 [95% CI, 0.58–1.72]), as well as the individual components of the primary composite outcome (Table 2; Table S11; Figure S10). At 30 days, pacemaker implantation remained significantly less frequent in the BE-THV group compared with the SE-THV group (unadjusted rates, 11.9% versus 18.6%; HR, 0.62, [95% CI, 0.47–0.83]; HR_{PSM}, 0.56 [95% CI, 0.35–0.90]; HR_{IPTW}, 0.57 [95% CI, 0.40–0.83]; HR_{MVA}, 0.55 [95% CI, 0.39–0.78]; Table 2; Figure 3; Table S11; Figure S11). There was no significant treatment-by-subgroup interaction in terms of 30-day pacemaker implantation between treatment groups and THV generation (Table S12). At 30 days, there were no significant crude and adjusted differences between THVs for all the other end points (Table 2; Table S11).

The 1-, 2-, and 3-Year Outcomes

Crude and adjusted estimates at 1, 2, and 3 years are reported in Table 3 and Table S13. At 3 years, consistent with 1- and 2-year analyses, death or stroke was not significantly different between BE-THV and SE-THV groups (unadjusted rates, 23.7% versus 26.2%; HR, 0.86 [95% CI, 0.65–1.13]; HR_{PSM}, 0.99 [95% CI, 0.65–1.51]; HR_{IPTW}, 0.89 [95% CI, 0.64–1.24]; HR_{MVA}, 0.94 [95% CI, 0.66–1.34]; Table 3; Figure 4; Table S13; Figure S12). At 1, 2, and 3 years, the adjusted estimates of the individual outcomes of death and stroke were not significantly different between THVs (Table 3; Table S13). No known ascending aorta rupture was observed at the maximum available follow-up. Pacemaker implantation across the different follow-up times remained significantly less frequent after BE-THV implantation than after SE-THV implantation (Table 3; Table S13). At 3 years, consistent with 1- and 2-year analyses, adjusted repeat aortic valve replacement and THV-related rehospitalization were not significantly different between THVs, regardless of the method used (Table 3; Table S13). Repeat TAVR at follow-up occurred almost invariably for severe aortic regurgitation. At the maximum available follow-up, there were no significant adjusted differences in the other outcomes (Table 3; Table S13). The assessment of outcomes trends within the period from the 1- to 3-year follow-up is illustrated in Table S14. Adjusted estimates did not reveal significant differences between treatment groups from the 1- to 3-year follow-up (Table S14). One-year echocardiography showed no significant adjusted difference in aortic valve mean gradient >20 mm Hg between the BE-THV and

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Table 1. Clinical, Anatomical, Functional, and Procedural Characteristics

	Total (n=1443)	BE-THV (n=860)	SE-THV (n=583)	*P* value	SMD	SMD_{MI}	SMD_{PSM}
Age, y	78.7 (72.6-83.2)	78.0 (71.3-82.5)	79.7 (74.1-84.0)	<0.001	−0.189	−0.189	−0.019
<75, n (%)	469 (32.5)	312 (36.3)	157 (26.9)				
≥75, n (%)	974 (67.5)	548 (63.7)	426 (73.1)				
Female, n (%)	569 (39.4)	299 (34.8)	270 (46.3)	<0.001	−0.242	−0.242	−0.002
Body mass index, kg/m^2	25.7 (23.1–29.3)	25.8 (23.3–29.4)	25.4 (22.8–29.1)	0.044	0.094	0.096	0.025
Diabetes, n (%)	346 (24.0)	215 (25.1)	131 (22.5)	0.266	0.059	0.059	−0.002
Hypertension, n (%)	1126 (78.1)	679 (79.1)	447 (76.7)	0.266	0.061	0.060	0.009
Coronary artery disease, n (%)	634 (44.1)	421 (49.3)	213 (36.5)	<0.001	0.255	0.253	0.029
Previous myocardial infarction, n (%)	187 (13.0)	111 (12.9)	76 (13.1)	0.933	−0.005	−0.004	0.011
Previous percutaneous coronary intervention, n (%)	369 (25.6)	240 (28.0)	129 (22.2)	0.014	0.129	0.129	0.002
Previous coronary artery bypass grafting, n (%)	136 (9.4)	84 (9.8)	52 (8.9)	0.588	0.029	0.029	0.027
Previous pacemaker, n (%)	133 (9.2)	77 (9.0)	56 (9.6)	0.690	−0.022	−0.020	0.015
Previous stroke, n (%)	138 (9.7)	89 (10.4)	49 (8.5)	0.229	0.063	0.060	0.011
Peripheral artery disease, n (%)	212 (15.0)	134 (15.8)	78 (13.6)	0.254	0.060	0.062	0.023
Chronic obstructive pulmonary disease, n (%)	263 (19.3)	151 (19.3)	112 (19.2)	0.967	0.002	0.017	−0.003
Estimated glomerular filtration rate, mL·min^{-1}·1.73 m^{-2}	62.5 (47.7–78.6)	63.0 (48.6–79.4)	62.0 (46.0–77.6)	0.122	0.082	0.083	0.015
≥60, n (%)	792 (55.3)	478 (55.8)	314 (54.4)				
30–59, n (%)	563 (39.3)	339 (39.6)	224 (38.8)				
<30, n (%)	78 (5.4)	39 (4.6)	39 (6.8)				
New York Heart Association class, n (%)				0.220			
0–2	485 (33.7)	294 (34.3)	191 (32.9)		0.030	0.030	0.009
3	809 (56.3)	469 (54.7)	340 (58.5)		−0.076	−0.074	−0.016
4	144 (10.0)	94 (11.0)	50 (8.6)		0.076	0.071	0.013
Left ventricular ejection fraction, %	56 (45–60)	55 (43–60)	57 (45–60)	0.020	−0.132	−0.133	0.010
≥50	982 (69.0)	576 (67.8)	406 (70.7)				
40–49	165 (11.6)	94 (11.1)	71 (12.4)				
30–39	135 (9.5)	88 (10.4)	47 (8.2)				
<30	142 (10.0)	92 (10.8)	50 (8.7)				
Aortic valve maximum gradient, mm Hg	76.0 (63.0–92.0)	76.0 (63.0–92.0)	76.5 (63.0–92.0)	0.889	−0.008		
Aortic valve mean gradient, mm Hg	46.0 (38.0–58.0)	46.0 (37.9–57.0)	47.0 (38.0–59.0)	0.269	−0.074	−0.070	0.001
Aortic valve area, cm^2	0.7 (0.6–0.8)	0.7 (0.6–0.8)	0.7 (0.5–0.8)	0.042	0.082	0.074	0.006
Indexed aortic valve area, cm^2/m^2	0.4 (0.3–0.4)	0.4 (0.3–0.4)	0.4 (0.3–0.4)	0.544	−0.033		
Aortic valve peak systolic velocity, m/s	4.3 (3.9–4.8)	4.3 (3.9–4.8)	4.3 (4.0–4.8)	0.447	−0.037	−0.013	−0.004
Aortic valve regurgitation, n (%)				0.364			
None or trivial	530 (37.7)	313 (37.3)	217 (38.4)		−0.024	0.042	0.007
Mild	616 (43.8)	373 (44.4)	243 (43.0)		0.028		
Moderate	199 (14.2)	124 (14.8)	75 (13.3)		0.042	0.008	−0.002
Severe	60 (4.3)	30 (3.6)	30 (5.3)		−0.094	−0.096	−0.010
Mitral valve regurgitation, n (%)				0.816			
None or trivial	326 (23.4)	193 (23.2)	133 (23.6)		−0.008	0.044	−0.005
Mild	744 (53.3)	437 (52.6)	307 (54.4)		−0.037		
Moderate	264 (18.9)	163 (19.6)	101 (17.9)		0.043	−0.051	0.004
Severe	61 (4.4)	38 (4.6)	23 (4.1)		0.024	0.021	0.000

(*Continued*)

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Table 1. Continued

	Total (n=1443)	BE-THV (n=860)	SE-THV (n=583)	*P* value	SMD	SMD_{MI}	SMD_{PSM}
Tricuspid valve regurgitation, n (%)				0.706			
None or trivial	329 (27.3)	208 (27.5)	121 (26.8)		0.017	0.051	0.019
Mild	720 (59.7)	447 (58.9)	275 (60.8)		−0.039		
Moderate	138 (11.4)	91 (12.1)	47 (10.4)		0.051	−0.031	−0.020
Severe	20 (1.7)	11 (1.5)	9 (2.0)		−0.045	−0.050	0.005
BAV type, n (%)				0.668			
0	136 (10.1)	80 (9.8)	56 (10.5)		−0.024	−0.073	0.001
1	1191 (88.4)	725 (88.8)	466 (87.6)		0.040	0.081	0.001
2	21 (1.6)	11 (1.3)	10 (1.9)		−0.046	−0.033	−0.004
BAV subtype, n (%)				0.403			
0–Lateral	100 (7.7)	57 (7.2)	43 (8.7)		−0.057		
0–Anteroposterior	30 (2.3)	18 (2.3)	12 (2.4)		−0.101		
1–Left-right	920 (71.2)	574 (72.2)	346 (69.6)		0.058		
1–Right-noncoronary	166 (12.8)	104 (13.1)	62 (12.5)		0.018		
1–Noncoronary-left	56 (4.3)	31 (3.9)	25 (5.0)		−0.058		
2–Left-right and right-noncoronary	12 (0.9)	6 (0.8)	6 (1.2)		−0.052		
2–Noncoronary-left and right-noncoronary	4 (0.3)	4 (0.5)	0 (0)		0.071		
2–Left-right and noncoronary-left	4 (0.3)	1 (0.1)	3 (0.6)		−0.135		
Annulus area, mm^2	500.0 (425.5–580.0)	520.0 (440.0–595.5)	470.0 (406.0–547.4)	<0.001	0.329	0.305	0.025
Annulus perimeter, mm	80.9 (74.4–87.2)	82.2 (75.8–88.4)	78.0 (72.5–84.3)	<0.001	0.336		
Maximum annular diameter, mm	28.2 (26.0–31.0)	29.0 (26.2–31.3)	27.6 (25.4–29.9)	<0.001	0.296		
Minimum annular diameter, mm	22.5 (20.7–24.3)	22.9 (21.0–24.8)	22.0 (20.0–23.8)	<0.001	0.354		
Raphe calcification, n (%)				<0.001			
None-mild	340 (31.4)	178 (26.2)	162 (40.2)		−0.318		
Moderate	433 (40.0)	287 (42.3)	146 (36.2)		0.122		
Severe	309 (28.6)	214 (31.5)	95 (23.6)		0.171		
Leaflet calcification, n (%)				<0.001			
None-mild	101 (10.0)	47 (7.5)	54 (14.1)		−0.250	−0.175	−0.018
Moderate	435 (43.3)	249 (40.0)	186 (48.7)		−0.178	−0.174	−0.013
Severe	469 (46.7)	327 (52.5)	142 (37.2)		0.307	0.268	0.022
Annulus calcification, n (%)				0.285			
None-mild	670 (60.5)	412 (60.6)	258 (60.4)		0.003	0.008	0.000
Moderate	316 (28.5)	201 (29.6)	115 (26.9)		0.058	0.051	−0.001
Severe	121 (10.9)	67 (9.9)	54 (12.6)		−0.094	−0.089	0.000
Left ventricular outflow calcification, n (%)				0.079			
None-mild	785 (71.4)	483 (70.8)	302 (72.2)		−0.031	−0.024	−0.004
Moderate	196 (17.8)	133 (19.5)	63 (15.1)		0.112	0.102	0.014
Severe	119 (10.8)	66 (9.7)	53 (12.7)		−0.102	−0.100	−0.013
Intercommissural distance, mm	28.5 (26.0–32.0)	29.7 (27.0–33.1)	27.0 (25.0–29.7)	<0.001	0.565		
Ascending aorta diameter, mm	37.0 (33.4–41.0)	37.5 (34.0–41.5)	36.5 (33.0–40.0)	0.002	0.158	0.146	0.021
<40, n (%)	915 (69.4)	533 (66.9)	382 (73.2)				
≥40, n (%)	404 (30.6)	264 (33.1)	140 (26.8)				
<45, n (%)	1212 (91.9)	729 (91.5)	483 (92.5)				
≥45, n (%)	107 (8.1)	68 (8.5)	39 (7.5)				
Sinotubular junction diameter	30.7 (28.0–33.6)	31.0 (28.1–34.0)	30.0 (27.3–33.0)	0.001	0.180		

(*Continued*)

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Table 1. Continued

	Total (n=1443)	BE-THV (n=860)	SE-THV (n=583)	*P* value	SMD	SMD_{MI}	SMD_{PSM}
Procedure period, n (%)				<0.001			
2007–2013	126 (8.8)	52 (6.1)	74 (12.7)		−0.277	−0.277	−0.015
2014–2017	712 (49.6)	470 (55.1)	242 (41.6)		0.272	0.272	0.051
2018–2021	597 (41.6)	331 (38.8)	266 (45.7)		−0.142	−0.142	−0.045
STS-PROM, %	3.2 (2.0–4.8)	3.1 (2.0–4.7)	3.2 (2.1–5.0)	0.350	−0.006	−0.001	0.005
Anesthesia, n (%)							
Local or conscious sedation	987 (71.3)	578 (69.9)	409 (73.3)	0.169	0.074	−0.078	−0.033
General	398 (28.7)	249 (30.1)	149 (26.7)				
Procedure timing, n (%)							
Elective	1393 (97.5)	822 (96.9)	571 (98.3)	0.111	0.078		
Emergency	36 (2.5)	26 (3.1)	10 (1.7)				
Access site, n (%)				0.001			
Femoral	1347 (93.3)	797 (92.7)	550 (94.3)		−0.064		
Transapical	42 (2.9)	36 (4.2)	6 (1.0)		0.158		
Transsubclavian	16 (1.1)	9 (1.0)	7 (1.2)		−0.015		
Transaortic	15 (1.0)	10 (1.2)	5 (0.9)		0.028		
Transcarotid	13 (0.9)	4 (0.5)	9 (1.5)		−0.159		
Transcaval	9 (0.6)	4 (0.5)	5 (0.9)		−0.058		
Other	1 (0.1)	0 (0)	1 (0.2)		−0.065		
Cerebral embolic protection device, n (%)	42 (3.2)	22 (2.8)	20 (3.8)	0.289	−0.064		
Predilation, n (%)	919 (65.0)	510 (60.1)	408 (72.5)	<0.001	−0.254		
THV generation, n (%)*							
Early	291 (20.2)	69 (8.0)	222 (38.1)	<0.001	1.106		
New	1152 (79.8)	791 (92.0)	361 (61.9)				
Postdilation, n (%)	406 (28.9)	155 (18.4)	251 (44.6)	<0.001	−0.674		
Contrast media volume, mL	140 (100–190)	130 (100–185)	146 (110–200)	0.002	−0.164		
Access site closure, n (%)				<0.001			
ProGlide	912 (70.4)	612 (77.3)	300 (59.5)		0.424		
Prostar	272 (21.0)	115 (14.5)	157 (31.2)		−0.472		
Manta	32 (2.5)	14 (1.8)	18 (3.6)		−0.137		
Surgical	62 (4.8)	39 (4.9)	23 (4.6)		0.017		
Multiple or other	18 (1.4)	12 (1.5)	6 (1.2)		0.027		

BAV indicates bicuspid aortic valve; BE-THV, balloon-expandable transcatheter heart valve; SE-THV, self-expandable transcatheter Heart Valve; SMD, standardized mean difference; SMD_{MI}, average standardized mean difference across the data sets generated by multiple imputation before propensity score matching; SMD_{PSM}, average standardized mean difference across the data sets generated by multiple imputation after propensity score matching; STS-PROM, Society of Thoracic Surgeons–Predicted Risk of Mortality; and THV, transcatheter heart valve.

Categorical variables are reported as counts and proportions. Continuous variables are reported as median and interquartile range. *P* values refer to formal testing of significant differences in baseline characteristics between treatment groups. SMD values refer to the assessment of baseline characteristics balance between treatment groups. SMD_{MI} values refer to the assessment of baseline characteristics balance between treatment groups across the 10 data sets generated by multiple imputation before PSM (average SMD_{MI} values). The minimum and maximum SMD_{MI} values across the 10 data sets generated by multiple imputation and the supplementary assessments by the Kolmogorov-Smirnov statistics and variance ratios are reported in Table S7 (left). SMD_{PSM} values refer to the assessment of baseline characteristics balance between treatment groups across the 10 data sets generated by multiple imputation after PSM (average SMD_{PSM} values). The minimum and maximum SMD_{PSM} values across the 10 data sets generated by multiple imputation and the supplementary assessments by the Kolmogorov-Smirnov statistics and variance ratios are reported in Table S7 (right). By convention, $P<0.05$ indicates a statistically significant difference. By convention, SMD, SMD_{MI}, and SMD_{PSM} values outside the range ±0.100 denote a significant between-group imbalance. Procedural variables with intrinsic correlation with the treatment (ie, access site, cerebral embolic protection device, predilation, THV generation, postdilation, contrast media volume, and access site closure) were not included in the model used to estimate the propensity scores, as recommended.[33] Nevertheless, major procedural characteristics were included in the multivariable models used for outcomes estimation according to the "doubly robust" adjustment.[38]

*Early-generation THV group includes Sapien XT, CoreValve, Acurate neo, Portico, Centera, and Allegra. New-generation THV group includes Sapien 3, Sapien 3 Ultra, Evolut R, Evolut Pro, and Evolut Pro+.

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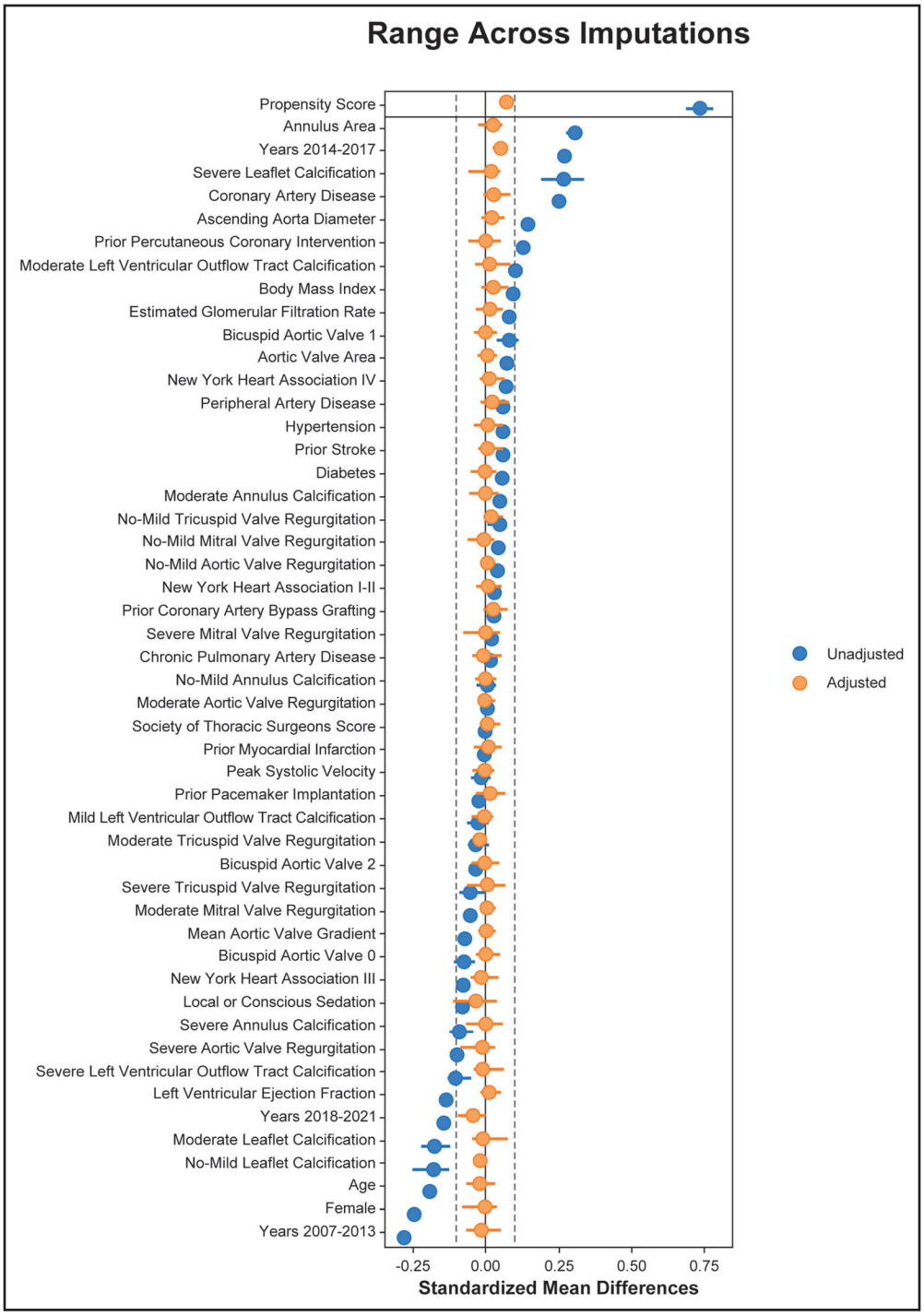

Figure 1. Balance between treatment groups before and after PSM.
Standardized mean differences (SMDs) across the 10 databases generated by multiple imputations before and after propensity score matching (PSM). Blue points represent SMDs in unadjusted estimates; orange points represent SMDs after PSM. After adjustment, the treatment groups achieved a good balance, as indicated by the orange points falling within the SMD range of ±0.100.[29] Although average estimates across the data sets were used to visually summarize the results, the "within-data set" method was used for data analysis because it showed superior performance compared with the "across–data set" method.[41] After PSM, the average adjusted sample size across imputed data sets was 497.0 patients in the SE-THV group and 497.0 in the BE-THV group.

SE-THV groups (Figure S13). In contrast, moderate to severe aortic regurgitation remained significantly less frequent in patients treated with BE-THV compared with those treated with SE-THV (5.6% versus 14.8%; OR_{MVA}, 0.29 [95% CI, 0.09–0.97]; Figure S14).

Subgroup and Sensitivity Analyses

The outcome of death or stroke at a 3-year follow-up was assessed across major subgroups (Figure 5; Figure S15). No significant treatment-by-subgroup

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Table 2. In-Hospital and 30-Day Outcomes

	Total (n=1443)	BE-THV (n=860)	SE-THV (n=583)	*P* value	OR/HR (95% CI)*†	OR_{PSM}/HR_{PSM} (95% CI)*†
In-hospital outcomes						
Death or stroke	67 (4.6)	34 (4.0)	33 (5.7)	0.132	0.69 (0.42–1.13)	0.93 (0.45–1.93)
Death	23 (1.6)	10 (1.2)	13 (2.2)	0.113	0.52 (0.22–1.18)	0.69 (0.18–2.69)
Stroke	47 (3.3)	26 (3.0)	21 (3.6)	0.549	0.84 (0.47–1.52)	1.09 (0.45–2.62)
Annulus rupture	14 (1.0)	11 (1.3)	3 (0.5)	0.145	2.51 (0.78–11.12)	6.85 (1.01–46.55)
Ventricular perforation	6 (0.4)	4 (0.5)	2 (0.3)	>0.999	1.36 (0.26–9.82)	2.03 (0.15–27.48)
Cardiac tamponade	15 (1.0)	10 (1.2)	5 (0.9)	0.575	1.36 (0.48–4.39)	3.11 (0.66–14.75)
Vascular complications	237 (16.4)	136 (15.8)	101 (17.2)	0.453		
Minor	167 (11.5)	97 (11.3)	70 (12.0)		0.92 (0.67–1.29)	0.92 (0.55–1.56)
Major	70 (4.9)	39 (4.5)	31 (5.3)		0.84 (0.52–1.37)	1.35 (0.55–3.27)
Coronary obstruction	8 (0.6)	4 (0.5)	4 (0.7)	0.722	0.68 (0.16–2.87)	0.63 (0.06–6.17)
Myocardial infarction	7 (0.5)	2 (0.2)	5 (0.9)	0.126	0.27 (0.04–1.26)	0.20 (0.01–3.09)
Additional THV implantation	42 (2.9)	11 (1.3)	31 (5.3)	<0.001	0.23 (0.11–0.45)	0.37 (0.13–1.00)
Intraoperative conversion to surgery	6 (0.4)	2 (0.2)	4 (0.7)	0.228	0.34 (0.05–1.73)	0.40 (0.04–4.19)
Pacemaker implantation‡	180 (13.8)	86 (11.0)	94 (17.8)	<0.001	0.57 (0.42–0.78)	0.55 (0.33–0.90)
Postprocedural aortic valve mean gradient, mm Hg	9.0 (7.0–13.0)	10.0 (8.0–13.0)	8.0 (6.0–11.0)	<0.001	1.97 (1.47–2.47)†	1.69 (0.88–2.50)†
Postprocedural aortic valve mean gradient grade, mm Hg				0.019		
<20	1258 (96.0)	749 (94.8)	509 (97.7)		Reference	Reference
20–29	50 (3.8)	38 (4.8)	12 (2.3)		2.15 (1.15–4.34)	2.38 (0.90–6.30)
30–39	3 (0.2)	3 (0.4)	0		4.76 (0.25–92.32)§	…
≥40	0	0	0		…	…
Postprocedural aortic valve regurgitation				<0.001		
None or trivial	756 (53.6)	509 (60.0)	247 (44.0)		Reference	Reference
Mild	552 (39.1)	311 (36.7)	241 (42.9)		0.63 (0.50–0.79)	0.51 (0.32–0.82)
Moderate	94 (6.7)	28 (3.3)	66 (11.7)		0.21 (0.13–0.33)	0.20 (0.08–0.48)
Severe	8 (0.6)	0 (0)	8 (1.4)		0.03 (0.00–0.50)§	…
Bleeding	248 (17.3)	154 (18.0)	94 (16.2)	0.375	1.14 (0.86–1.51)	1.26 (0.77–2.08)
Acute kidney injury	81 (5.6)	39 (4.5)	42 (7.2)	0.032	0.61 (0.39–0.96)	0.64 (0.27–1.51)
Postprocedural length of stay, d	5 (3–7)	5 (3–7)	6 (4–9)	<0.001	−1.53 (−2.24 to −0.81)†	−0.66 (−1.80 to 0.48)†
30-d Outcomes						
Death or stroke	78 (5.5)	43 (5.1)	35 (6.1)	0.404	0.83 (0.53–1.29)	1.02 (0.51–2.02)
Death	31 (2.2)	16 (1.9)	15 (2.7)	0.350	0.72 (0.35–1.45)	0.82 (0.27–2.46)
Cardiac death	21 (1.5)	9 (1.1)	12 (2.1)	0.115	0.51 (0.21–1.20)	0.55 (0.14–2.13)
Stroke	51 (3.6)	29 (3.4)	22 (3.8)	0.692	0.89 (0.51–1.56)	1.18 (0.51–2.77)
Repeat valve replacement	10 (0.8)	3 (0.4)	7 (1.3)	0.046	0.28 (0.07–1.07)	0.12 (0.00–4.24)
Valve thrombosis	4 (0.3)	3 (0.4)	1 (0.2)	0.559	1.94 (0.20–18.63)	1.91 (0.14–25.62)
Endocarditis	2 (0.2)	2 (0.3)	0	0.244	3.40 (0.08–140.00)‖	…
Myocardial infarction	10 (0.7)	5 (0.6)	5 (0.9)	0.555	0.69 (0.20–2.38)	0.67 (0.09–5.23)
Pacemaker implantation‡	185 (14.6)	90 (11.9)	95 (18.6)	0.001	0.62 (0.47–0.83)	0.56 (0.36–0.88)
Valve-related rehospitalization	23 (1.9)	11 (1.5)	12 (2.4)	0.272	0.64 (0.28–1.44)	0.65 (0.13–3.13)

BE-THV indicates balloon-expandable transcatheter aortic valve; HR, hazard ratio; OR, odds ratio; PSM, propensity score matching; SE-THV, self-expandable transcatheter aortic valve; and THV, transcatheter heart valve.

Values are counts and proportions for in-hospital outcomes and counts and Kaplan-Meier estimates for 30-day outcomes.

*ORs refer to in-hospital outcomes and HRs refer to 30-day outcomes.

†Coefficients and 95% CIs by linear regression for the continuous outcomes of postprocedural mean gradient and postprocedural length of stay. Linear mixed-effects models were used to compute adjusted estimates after PSM.

‡Patients with previous pacemaker implantation were excluded.

§OR and 95% CI were computed by logistic regression with the Firth correction.

‖HR and 95% CI were computed by Cox regression with the Firth correction.

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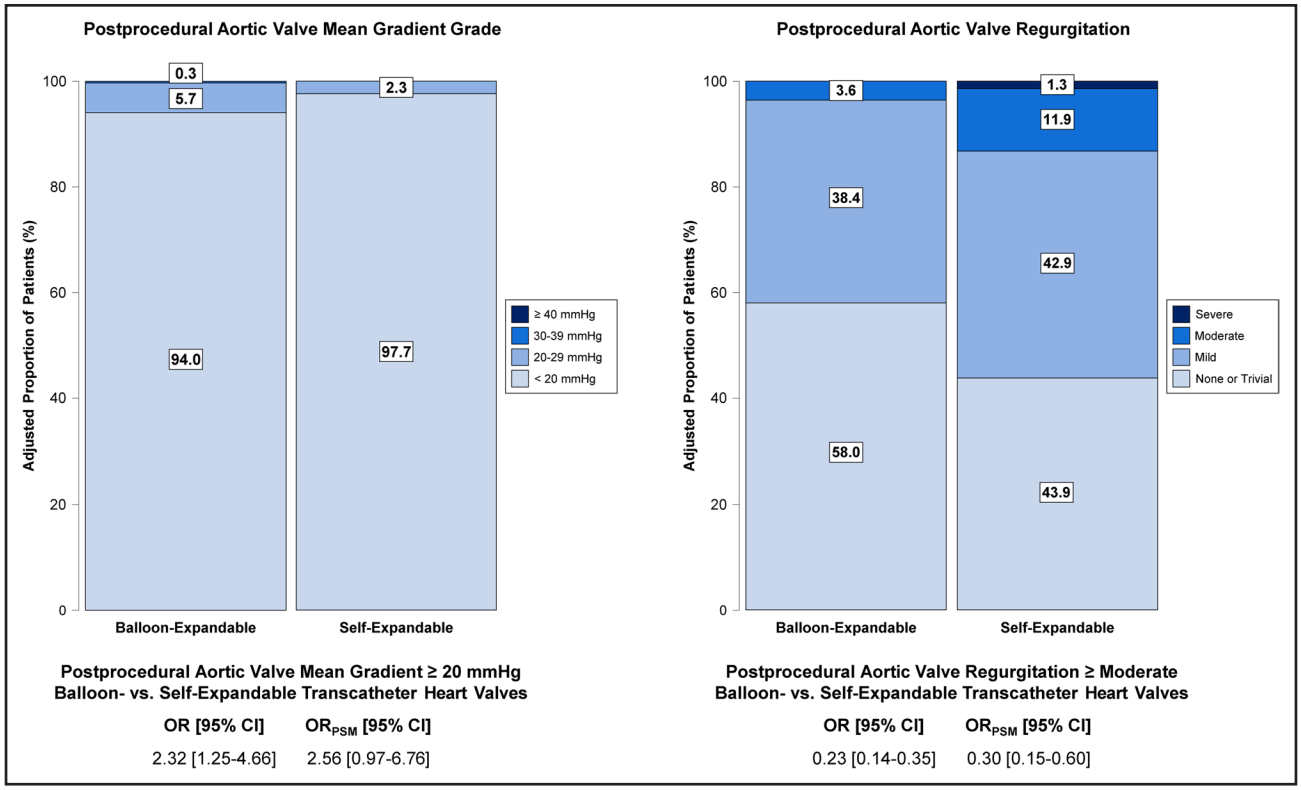

Figure 2. Postprocedural aortic valve mean gradient and postprocedural aortic valve regurgitation.
Average adjusted proportions of postprocedural mean gradient (**left**) and postprocedural regurgitation (**right**) between treatment groups after PSM. OR indicates odds ratio; and PSM, propensity score matching.

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interaction was observed, regardless of the adjustment method used. A sensitivity analysis with the exclusion of patients without an assessable BAV type showed consistent results (Table S15). A sensitivity analysis focusing on patients with BAV type 1 or 2 characterized by moderate to severe raphe(s) and leaflet calcification showed mitigated differences between groups in terms of pacemaker implantation and a possible late benefit in terms of stroke associated with BE-THV implantation compared with SE-THV implantation (Table S16). The comparison between patients who received a BE-THV and those who received an SE-THV by including only the most implanted types of THV did not reveal inconsistency (Table S17). The assessment of associations between major TAVR complications and computed tomography, echocardiography, and procedural characteristics is reported in Table S18. The analysis confirmed that BE-THV or SE-THV use was significantly associated with annulus rupture, postprocedural aortic mean gradient ≥20 mm Hg, and moderate to severe aortic regurgitation.

DISCUSSION

The main result of this study of patients undergoing TAVR for BAV stenosis is that the composite outcome of death or stroke does not significantly differ between patients who received a BE-THV and those who received an SE-THV at short- and long-term follow-up. Nevertheless, BE-THV implantation is more frequently associated with significant postprocedural aortic mean gradient and annulus rupture compared with SE-THV implantation. Moreover, the need for permanent pacemaker implantation is substantially lower after BE-THV implantation than after SE-THV implantation. In contrast, SE-THV implantation is more frequently associated with significant postprocedural aortic regurgitation and additional THV implantation compared with BE-THV implantation. Of note, postprocedural functional differences between the BE-THV and SE-THV groups seem to be consistent between early- and new-generation THVs. Overall, major clinical, anatomical, functional, and procedural conditions do not seem to significantly influence the results in death or stroke between the BE-THV and SE-THV groups. However, the presence of moderate to severe raphe(s) and leaflet calcification may be associated with some variations in long-term outcomes.

TAVR in patients with BAV stenosis can pose procedural challenges caused by unfavorable anatomical features that can include elliptically shaped, noncircular, and enlarged annular circumference; calcified raphe; bulky and asymmetrical calcifications of the leaflets with extension to the left ventricular outflow tract and aortic root; hostile supra-annular geometry; and dilated ascending

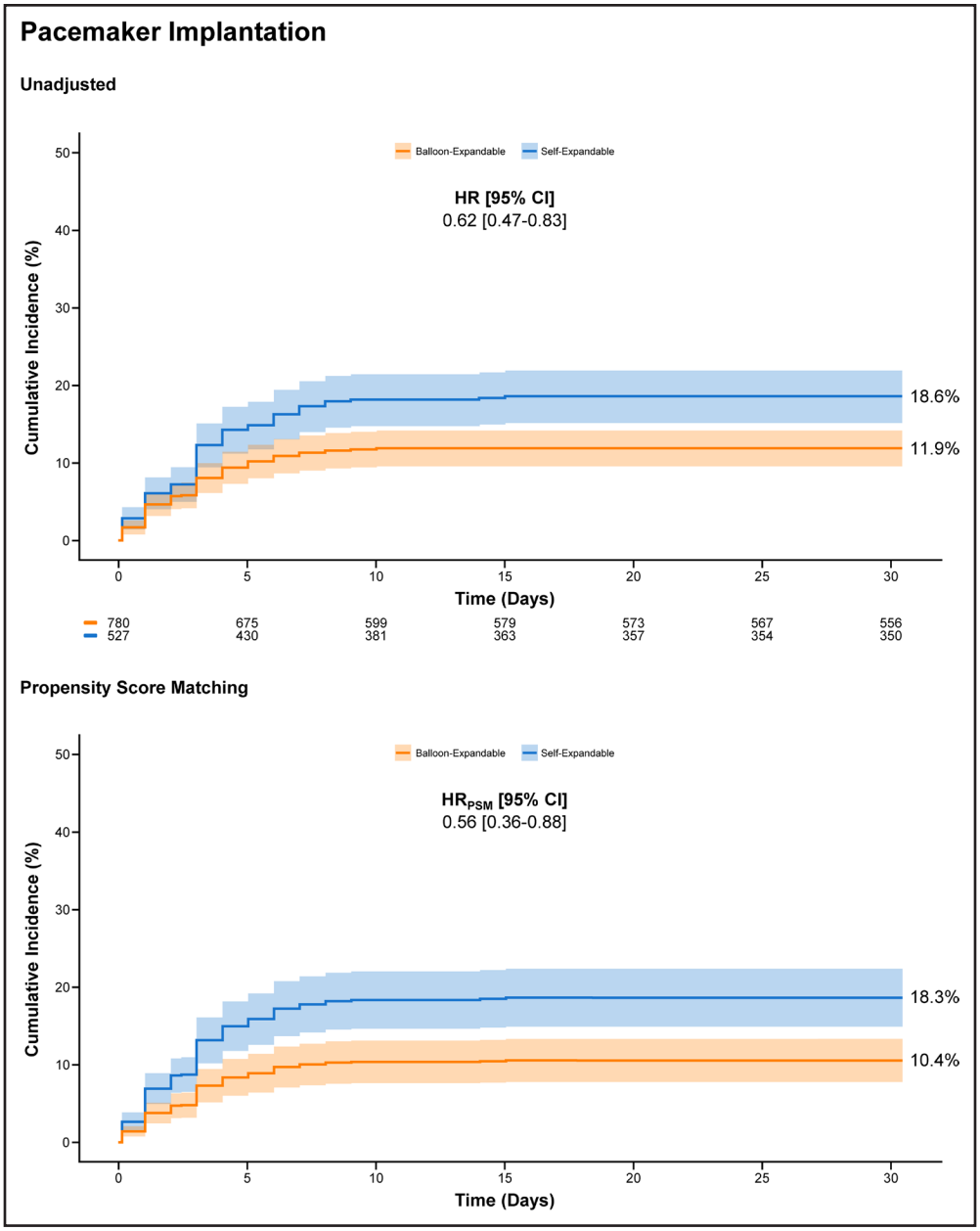

Figure 3. Permanent pacemaker implantation at short-term follow-up.
HR indicates hazard ratio; and PSM, propensity score matching.

aorta.[3,4,7,8,24,25,50,51] PSM-based registries and study-level meta-analyses comparing outcomes after TAVR between BAV and tricuspid aortic valve showed that BAV is associated with a higher risk of annular rupture, additional THV implantation, postprocedural moderate to severe aortic valve regurgitation, conversion to surgery, in-hospital stroke, and permanent pacemaker implantation.[21,22,52,53] Against this background, it is plausible that differences between THV deployment methods and related differences in technical features can influence the outcomes of TAVR in patients with BAV stenosis.[5,24,25] Recently, an observational study focused for the first time on the comparison between patients who received a BE-THV and patients who received an SE-THV for BAV stenosis.[26] However, the study could not adequately assess clinical outcomes because results were based on 77 matched pairs of patients and the maximum follow-up was 1 year.[26] In the absence of randomized clinical trials and large registries directly comparing BE-THV with SE-THV implantation in patients with BAV stenosis, the present study addresses differences between the 2 main THV technologies in a substantial number of patients.

In our study, in-hospital death or stroke between the BE-THV and SE-THV groups and the individual components of the composite outcome were not significantly different. Nevertheless, BE-THV implantation was

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Table 3. Long-Term Outcomes

	Total (n=1443)	BE-THV (n=860)	SE-THV (n=583)	*P* value	HR (95% CI)	HR_{PSM} (95% CI)
1-y Outcomes						
Death or stroke	140 (11.1)	81 (10.7)	59 (11.5)	0.570	0.91 (0.65–1.27)	1.01 (0.61–1.70)
Death	90 (7.6)	54 (7.6)	36 (7.4)	0.933	0.98 (0.65–1.50)	1.02 (0.51–2.04)
Cardiac death	46 (3.7)	26 (3.6)	20 (3.9)	0.614	0.86 (0.48–1.54)	0.99 (0.38–2.56)
Stroke	60 (4.5)	31 (3.7)	29 (5.6)	0.205	0.72 (0.44–1.20)	0.88 (0.40–1.95)
Repeat valve replacement	16 (1.4)	4 (0.6)	12 (2.6)	0.003	0.21 (0.07–0.66)	0.37 (0.06–2.41)
Valve thrombosis	7 (0.6)	6 (0.9)	1 (0.2)	0.181	3.83 (0.46–31.79)	3.45 (0.22–53.79)
Endocarditis	7 (0.7)	5 (0.8)	2 (0.5)	0.544	1.65 (0.32–8.52)	0.85 (0.07–11.17)
Myocardial infarction	13 (1.0)	7 (1.0)	6 (1.1)	0.685	0.80 (0.27–2.38)	0.49 (0.03–7.49)
Pacemaker implantation*	202 (16.6)	100 (13.9)	102 (20.7)	0.001	0.64 (0.49–0.84)	0.55 (0.36–0.85)
Valve-related rehospitalization	68 (6.4)	27 (4.2)	41 (9.6)	<0.001	0.44 (0.27–0.72)	0.58 (0.17–1.90)
2-y Outcomes						
Death or stroke	180 (17.4)	101 (15.9)	79 (19.8)	0.190	0.82 (0.61–1.10)	1.00 (0.65–1.56)
Death	129 (13.7)	75 (13.0)	54 (14.9)	0.473	0.88 (0.62–1.25)	1.09 (0.63–1.90)
Cardiac death	67 (7.3)	34 (5.9)	33 (9.6)	0.083	0.66 (0.41–1.06)	0.90 (0.39–2.06)
Stroke	65 (5.4)	31 (3.7)	34 (8.1)	0.044	0.61 (0.38–0.99)	0.73 (0.35–1.51)
Repeat valve replacement	19 (1.9)	6 (1.1)	13 (3.1)	0.007	0.29 (0.11–0.76)	0.49 (0.02–9.75)
Valve thrombosis	9 (0.9)	8 (1.4)	1 (0.2)	0.093	4.97 (0.62–39.74)	3.95 (0.30–52.19)
Endocarditis	10 (1.2)	7 (1.3)	3 (0.9)	0.560	1.49 (0.39–5.77)[c]	0.87 (0.07–10.64)
Myocardial infarction	16 (1.6)	8 (1.3)	8 (2.2)	0.407	0.66 (0.25–1.77)	0.21 (0.03–1.65)
Pacemaker implantation*	211 (18.3)	104 (15.1)	107 (23.2)	0.001	0.63 (0.48–0.82)	0.55 (0.36–0.83)
Valve-related rehospitalization	85 (9.2)	35 (6.5)	50 (13.4)	<0.001	0.45 (0.29–0.70)	0.63 (0.23–1.72)
3-y Outcomes						
Death or stroke	208 (24.7)	120 (23.7)	88 (26.2)	0.287	0.86 (0.65–1.13)	0.99 (0.65–1.51)
Death	158 (21.2)	94 (20.8)	64 (22.0)	0.585	0.92 (0.67–1.26)	1.07 (0.63–1.80)
Cardiac death	73 (9.0)	37 (7.1)	36 (12.1)	0.064	0.65 (0.41–1.03)	0.67 (0.18–2.49)
Stroke	66 (5.7)	32 (4.2)	34 (8.1)	0.056	0.63 (0.39–1.02)	0.75 (0.37–1.51)
Repeat valve replacement	19 (1.9)	6 (1.1)	13 (3.1)	0.007	0.29 (0.11–0.76)	0.49 (0.02–9.75)
Valve thrombosis	11 (1.6)	10 (2.4)	1 (0.2)	0.051	6.03 (0.77–47.17)	4.65 (0.30–73.31)
Endocarditis	11 (1.4)	7 (1.3)	4 (1.7)	0.865	1.11 (0.33–3.80)	0.58 (0.05–7.19)
Myocardial infarction	20 (2.8)	10 (2.1)	10 (4.0)	0.324	0.65 (0.27–1.55)	0.33 (0.06–1.70)
Pacemaker implantation*	214 (19.3)	105 (15.5)	109 (25.1)	<0.001	0.62 (0.48–0.81)	0.53 (0.35–0.81)
Valve-related rehospitalization	91 (11.1)	38 (7.9)	53 (16.1)	<0.001	0.46 (0.30–0.69)	0.64 (0.25–1.69)

BE-THV indicates balloon-expandable transcatheter aortic valve; HR, hazard ratio; PSM, propensity score matching; and SE-THV, self-expandable transcatheter aortic valve.

Values are counts and Kaplan-Meier estimates at the 1-, 2-, and 3-year follow-up.

*Patients with previous pacemaker implantation were excluded.

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associated with an increased rate of the life-threatening complication of annulus rupture compared with SE-THV implantation, which in the present study occurred 5 to 6 times more frequently than in studies of patients with tricuspid aortic valve stenosis treated with BE-THV.[11,13,15] In contrast, SE-THV implantation was associated with a higher rate of additional THV implantation compared with BE-THV implantation (average absolute rate difference, 3.8%). We also found some functional acute differences between the BE-THV and SE-THV groups. Indeed, BE-THV implantation was more frequently associated with postprocedural mean gradient ≥20 mm Hg, whereas SE-THV implantation was associated more frequently with moderate to severe postprocedural regurgitation. Of note, a sensitivity analysis according to THV generation indicated that, regardless of the adjustment method, differences in postprocedural mean gradient and postprocedural regurgitation between BE-THV and SE-THV

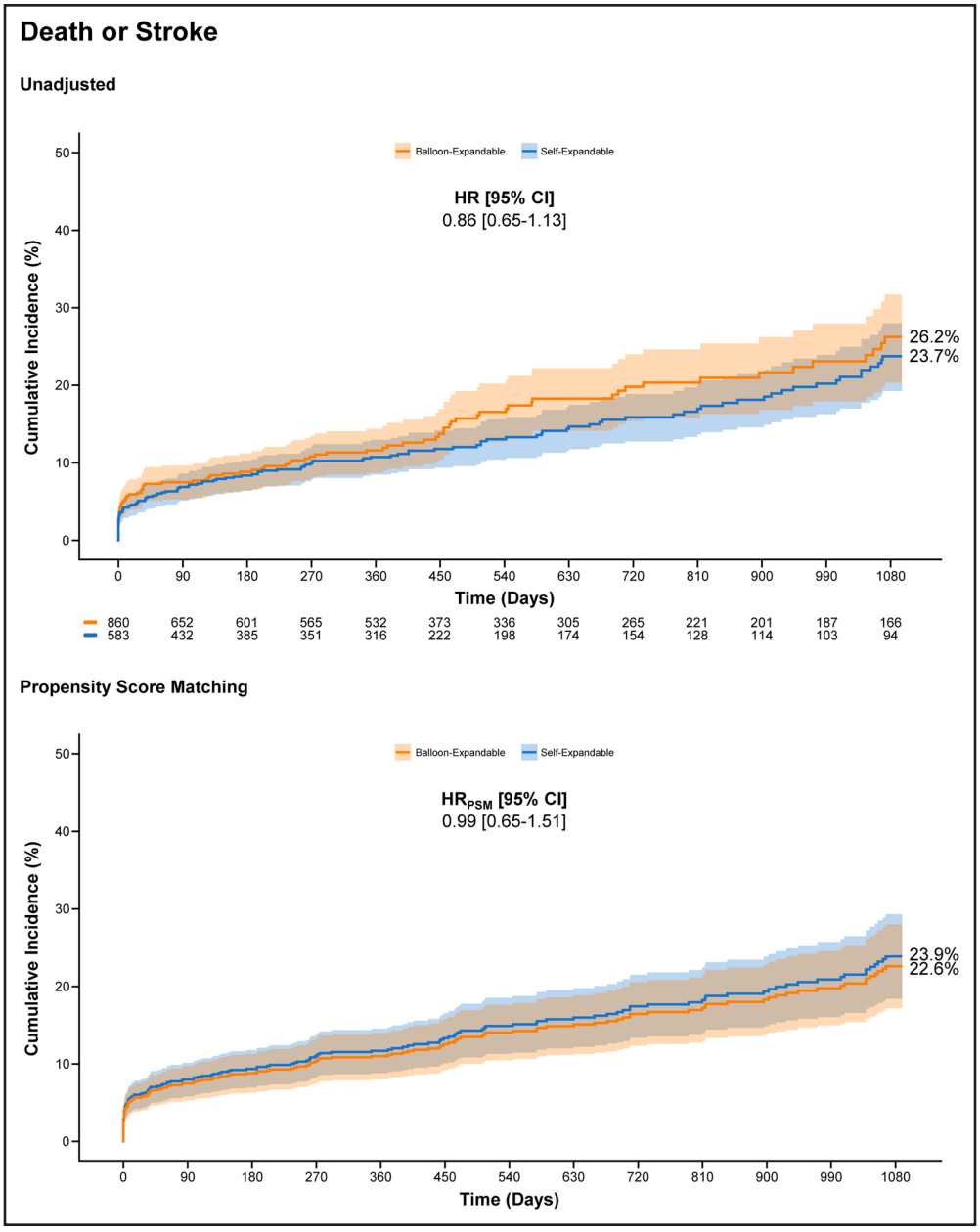

Figure 4. Death or stroke or death at long-term follow-up.
HR indicates hazard ratio; and PSM, propensity score matching.

within the early- and new-generation subsets were consistent. In the only available report from the STS/ACC TVT registry providing limited information on the comparison between patients who received a BE-THV and those who received an SE-THV for the treatment of BAV stenosis, Halim et al[23] reported an increased occurrence of post-TAVR moderate to severe aortic regurgitation with SE-THV compared with BE-THV despite the use of current-generation THVs (adjusted OR, 2.97 [95% CI, 1.93–4.59]). This previous finding is consistent with the results of the present study.

The absence of significant differences in death or stroke between BE-THV and SE-THV groups across different time points up to the 3-year follow-up is the major conclusion of this study. Of note, after analyzing the individual components of death and stroke at the conventional time points of 30 days and 1 year, we did not detect significant differences between patients who received a BE-THV and those who received an SE-THV, and results were consistent after adjustment. Comparing the incidences of death at 30 days and 1 year in the present study with those reported in a limited number of relevant studies independently evaluating BE-THV or SE-THV implantation for BAV stenosis against the same type of THV used to treat tricuspid aortic valve stenosis poses challenges because of the heterogeneous baseline risk of the patients.[18,21,22,54,55] Indeed, a few available studies from the STS/ACC TVT registry included

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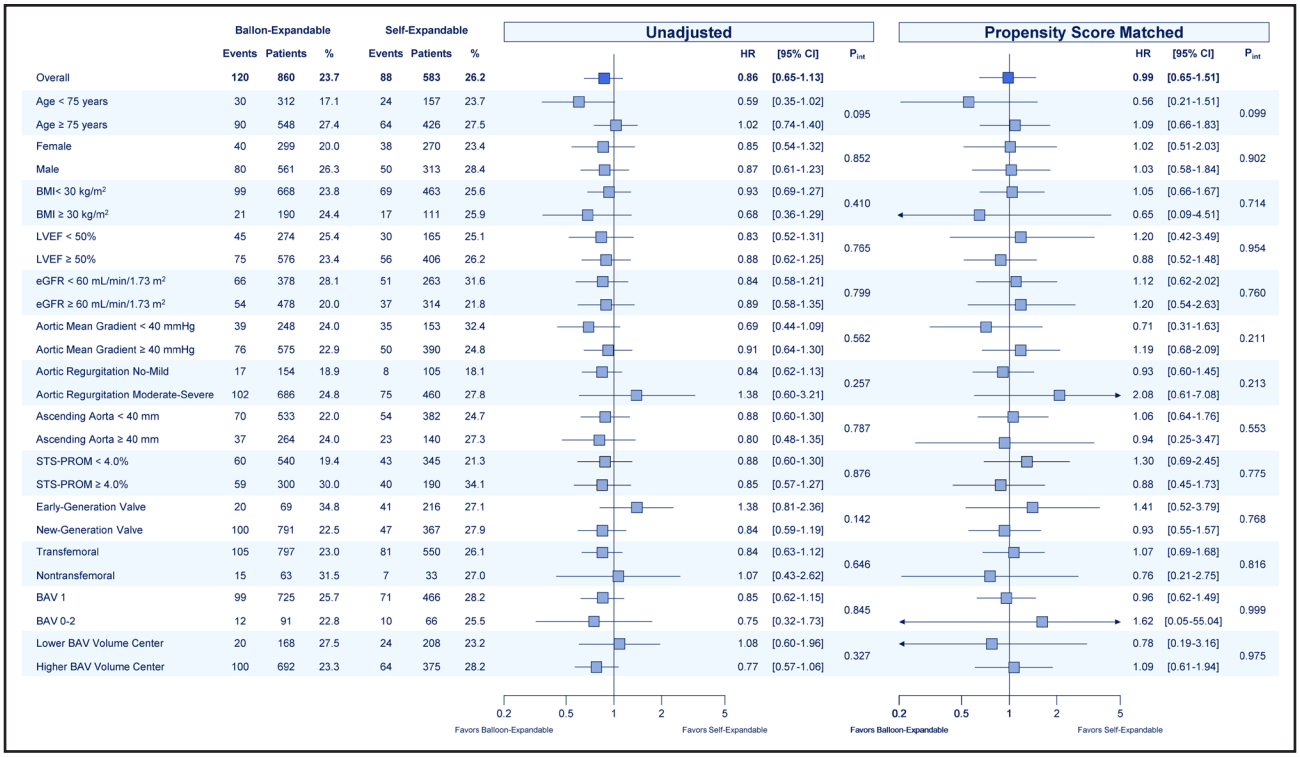

Figure 5. Subgroup analyses for death or stroke at long-term follow-up.
BAV indicates bicuspid aortic valve; BMI, body mass index; eGFR, estimated glomerular filtration rate; HR, hazard ratio; LVEF, left ventricular ejection fraction; P_{int}, $P_{interaction}$; PSM, propensity score matching; and STS-PROM, Society of Thoracic Surgeons–Predicted Risk of Mortality.

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patients with a more complex individual risk profile (mean STS-PROM, 4.9%–5.1%) compared with those included in the present study (mean STS-PROM, 4.1%; median STS-PROM, 3.2%).[21,22] Conversely, reports on TAVR with BAV in the low-risk setting included selected patients with a less complex individual risk profile (mean STS-PROM, 1.4%–1.7%) compared with those included in our study (mean STS-PROM, 4.1%; median STS-PROM, 3.2%).[18,54,55] Against this background, we found adjusted incidences of death (30 days: BE-THV, 2.0% to 2.3% and SE-THV, 2.1% to 2.2% across adjustment methods; 1 year: BE-THV, 7.1% to 7.5% and SE-THV, 6.6% to 7.0% across adjustment methods) that were lower than those observed in more complex populations by Makkar and colleagues[21] (BE-THV: 30-day and 1-year death, 2.6% and 10.5%) and Forrest and colleagues[22] (SE-THV: 30-day and 1-year death, 2.6% and 10.4%) and higher than those observed in low-risk patients by Makkar and colleagues[18] (BE-THV; 30-day and 1-year death, 0.9% and 4.6%) and Deeb and colleagues[54,55] (SE-THV; 30-day and 1-year death, 0.7% and 0.7%). The adjusted incidence of stroke in our study (30-day and 1-year stroke: BE-THV, 2.9%–3.2% and 3.7%–4.0% across methods; SE-THV, 4.0%–4.1% and 5.0%–5.2% across methods) was higher compared with both high-risk (BE-THV: 30-day and 1-year stroke, 2.5% and 3.4%; SE-THV: 30-day and 1-year stroke, 3.4% and 3.9%) and low-risk (BE-THV: 30-day and 1-year stroke, 1.4% and 2.0%; SE-THV: 30 day and 1-year death and stroke, 4.0% and 4.8%) patients.[18,21,22,54,55]

Our findings also contribute valuable information on clinical outcomes of patients undergoing TAVR for BAV stenosis because available evidence to date was very limited and, to the best of our knowledge, essentially restricted to a maximum follow-up of 1 year, with only a registry reporting unadjusted 2-year outcomes according to BAV features.[5] In our study, 3-year adjusted rates of death or stroke after MVA, IPTW, and PSM ranged from 20.8% to 22.1% in patients who received a BE-THV and from 20.7% to 22.5% in patients who received an SE-THV. These unsatisfactory figures may have different concurrent explanations. On the one hand, these results likely reflect the higher anatomical complexity of the BAV compared with the tricuspid aortic valve. Yoon and colleagues[56] showed that substantial BAV leaflet calcification was associated with higher mortality after TAVR and reported unadjusted 2-year incidences (18.9%) that were significantly higher than those reported in our study (13.7%). Of note, in the study by Yoon and colleagues, no adjustment for potential confounders was used, the analysis essentially focused on mortality, and included a limited number of patients after stratification according to the calcification patterns.[56] In our study, sensitivity analyses including patients with moderate to severe raphe calcification combined with moderate to severe leaflet calcification showed no significant improvement in

the need for permanent pacing associated with BE-THV implantation compared with SE-THV implantation and a late signal of stroke reduction. These findings are challenging to interpret and may merely reflect the effect of reduced statistical power and a play of chance, respectively. On the other hand, it should also be acknowledged that differences between real-world study populations may derive from the inclusion of variable proportions of higher-risk patients who were referred to TAVR despite surgery being the recommended treatment because of unfavorable extracardiac conditions with an influence on long-term survival and other major outcomes.

Although we acknowledge the multifactorial influence on clinical outcomes of some BAV-specific clinical and anatomical features, the absence of significant adjusted differences between BE-THV and SE-THV and the high consistency of subgroup analyses support the similar safety and efficacy between BE-THV and SE-THV in patients undergoing TAVR for BAV stenosis. Specifically, the death or stroke between the BE-THV and SE-THV groups across key clinical (age, sex, body mass index), functional (left ventricular dysfunction, chronic kidney disease, preprocedural aortic valve mean gradient, preprocedural aortic valve regurgitation), and procedural (THV generation, access site) settings with established prognostic value and remarkable implications for the heart valve team were consistent with the main analysis. Similarly, 2 established factors with relevant impact on the outcomes of TAVR for BAV, ascending aorta diameter and STS-PROM, did not significantly influence the outcomes between the BE-THV and SE-THV groups. Consistently, there was no significant interaction between centers providing a larger number of patients and those providing a smaller number of patients. In addition, the restriction of the analysis to the most commonly implanted types of THVs, prompted by the recent approval in the United States and Europe for the treatment of selected patients with BAV stenosis, did not reveal inconsistency.

Finally, another major conclusion of the study is the observation of substantial differences in the need for permanent pacemaker implantation between patients who received a BE-THV and those who received an SE-THV. Available data in the tricuspid aortic valve setting showed heterogeneous rates of pacemaker implantation after TAVR, but in general, the occurrence is lower after BE-THV implantation compared with SE-THV implantation.[57] Our study confirms in the setting of BAV stenosis lower incidences of pacemaker implantation after BE-THV than SE-THV for an adjusted 30-day relative risk reduction of 43% to 45% and cumulative rates of 9.4% to 11.4% in the BE-THV group and 18.5% to 19.4% in the SE-THV group. Of note, the difference in pacemaker implantation between THVs remained unchanged across follow-up times.

Limitations

Some limitations need to be considered in the interpretation of the results of this study. First, the observational design is susceptible to confounders; the decision to implant a BE-THV or an SE-THV could be influenced by clinical and anatomical considerations. In addition, certain center- and operator-related influences may have played a role. However, we implemented rigorous and conservative methods to adjust the outcomes, including PSM with "doubly robust" adjustment, IPTW with doubly robust adjustment, and MVA, and we systematically accounted for center-specific factors by mixed-effects models. Second, some key settings may have exerted a higher influence on the primary outcomes, and our results may not be generalizable to any subgroup of patients with BAV stenosis. However, we performed multiple sensitivity and subgroup analyses, which generally yielded consistent results. Third, in our study, 20.1% of patients received early-generation THVs. Nevertheless, our assessment of short-term TAVR performance, 30-day pacemaker implantation, and 3-year death or stroke by THV generation did not reveal significant treatment-by-subgroup interaction, regardless of the adjustment method used. Fourth, in this study, echocardiography and computed tomography assessments were not based on a core laboratory, and clinical outcomes were site reported. Currently, however, no randomized trial on TAVR for BAV stenosis is available, and the highest-level evidence on the topic stems from administrative registries, wherein estimates not only are based on self-reporting but also lack the possibility of verification against the sources. The results of administrative registries rely on code-based identification strategies with known inherent limitations, primarily low discrimination and poor granularity. Some reports indicate that the aforementioned concerns, contingent on the object of interest, may be exacerbated by both provider misuse and medical uncertainty.[58] In contrast, the participating centers of this study shared an established program associated with the prospective collection of clinical, laboratory, instrumental, and procedural data, as well as clinical outcomes. In addition, systematic reviews of data from randomized trials generally did not show significant differences between site-reported and centrally adjudicated outcomes.[59] Last, computed tomography scan information was not readable or retrievable in 95 patients (6.6%) for multiple reasons. However, the influence of this limitation is negligible given that patients with irretrievable scan information still had BAV anatomy as evaluated in the original computed tomography assessment. In addition, multiple imputation techniques were used to partially overcome the absence of information in multivariable models, and sensitivity analyses excluding patients without available BAV type did not uncover inconsistency.

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Conclusions

In patients undergoing TAVR for BAV, death or stroke does not significantly differ between patients who received a BE-THV and those who received an SE-THV over a follow-up of 3 years. BE-THV is associated with a higher transvalvular mean gradient and a higher risk of annulus rupture. In contrast, SE-THV is associated with more frequent moderate to severe aortic regurgitation and a higher risk of additional THV implantation. Treatment of BAV stenosis with a BE-THV is associated with significantly lower pacemaker implantation than treatment with an SE-THV.

ARTICLE INFORMATION

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Supplemental Material

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APPENDIX

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REFERENCES

1. Hoffman JI, Kaplan S. The incidence of congenital heart disease. *J Am Coll Cardiol*. 2002;39:1890–1900. doi: 10.1016/s0735-1097(02)01886-7
2. Siu SC, Silversides CK. Bicuspid aortic valve disease. *J Am Coll Cardiol*. 2010;55:2789–2800. doi: 10.1016/j.jacc.2009.12.068
3. Michelena HI, Prakash SK, Della Corte A, Bissell MM, Anavekar N, Mathieu P, Bossé Y, Limongelli G, Bossone E, Benson DW, et al; BAVCon Investigators. Bicuspid aortic valve: identifying knowledge gaps and rising to the challenge from the International Bicuspid Aortic Valve Consortium (BAVCon). *Circulation*. 2014;129:2691–2704. doi: 10.1161/CIRCULATIONAHA.113.007851
4. Michelena HI, Della Corte A, Evangelista A, Maleszewski JJ, Edwards WD, Roman MJ, Devereux RB, Fernández B, Asch FM, Barker AJ, et al. International consensus statement on nomenclature and classification of the congenital bicuspid aortic valve and its aortopathy, for clinical, surgical, interventional and research purposes. *Eur J Cardiothorac Surg*. 2021;60:448–476. doi: 10.1093/ejcts/ezab038
5. Vincent F, Ternacle J, Denimal T, Shen M, Redfors B, Delhaye C, Simonato M, Debry N, Verdier B, Shahim B, et al. Transcatheter aortic valve replacement in bicuspid aortic valve stenosis. *Circulation*. 2021;143:1043–1061. doi: 10.1161/CIRCULATIONAHA.120.048048
6. Michelena HI, Desjardins VA, Avierinos JF, Russo A, Nkomo VT, Sundt TM, Pellikka PA, Tajik AJ, Enriquez-Sarano M. Natural history of asymptomatic patients with normally functioning or minimally dysfunctional bicuspid aortic valve in the community. *Circulation*. 2008;117:2776–2784. doi: 10.1161/CIRCULATIONAHA.107.740878
7. Roberts WC, Ko JM. Frequency by decades of unicuspid, bicuspid, and tricuspid aortic valves in adults having isolated aortic valve replacement for aortic stenosis, with or without associated aortic regurgitation. *Circulation*. 2005;111:920–925. doi: 10.1161/01.CIR.0000155623.48408.C5
8. Roberts WC, Ko JM, Hamilton C. Comparison of valve structure, valve weight, and severity of the valve obstruction in 1849 patients having isolated aortic valve replacement for aortic valve stenosis (with or without associated aortic regurgitation) studied at 3 different medical centers in 2 different time periods. *Circulation*. 2005;112:3919–3929. doi: 10.1161/CIRCULATIONAHA.105.543280
9. Otto CM, Nishimura RA, Bonow RO, Carabello BA, Erwin JP, Gentile F, Jneid H, Krieger EV, Mack M, McLeod C, et al; Writing Committee Members. 2020 ACC/AHA guideline for the management of patients with valvular heart disease. *J Am Coll Cardiol*. 2021;77:e25–e197. doi: 10.1016/j.jacc.2020.11.018
10. Vahanian A, Beyersdorf F, Praz F, Milojevic M, Baldus S, Bauersachs J, Capodanno D, Conradi L, De Bonis M, De Paulis R, et al. 2021 ESC/EACTS guidelines for the management of valvular heart disease. *EuroIntervention*. 2022;17:e1126–e1196. doi: 10.4244/EIJ-E-21-00009
11. Smith CR, Leon MB, Mack MJ, Miller DC, Moses JW, Svensson LG, Tuzcu EM, Webb JG, Fontana GP, Makkar RR, et al; PARTNER Trial Investigators. Transcatheter versus surgical aortic-valve replacement in high-risk patients. *N Engl J Med*. 2011;364:2187–2198. doi: 10.1056/NEJMoa1103510
12. Adams DH, Popma JJ, Reardon MJ, Yakubov SJ, Coselli JS, Deeb GM, Gleason TG, Buchbinder M, Hermiller J, Kleiman NS, et al; U.S. CoreValve Clinical Investigators. Transcatheter aortic-valve replacement with a self-expanding prosthesis. *N Engl J Med*. 2014;370:1790–1798. doi: 10.1056/NEJMoa1400590
13. Leon MB, Smith CR, Mack MJ, Makkar RR, Svensson LG, Kodali SK, Thourani VH, Tuzcu EM, Miller DC, Herrmann HC, et al; PARTNER 2 Investigators. Transcatheter or surgical aortic-valve replacement in intermediate-risk patients. *N Engl J Med*. 2016;374:1609–1620. doi: 10.1056/NEJMoa1514616
14. Reardon MJ, van Mieghem NM, Popma JJ, Kleiman NS, Søndergaard L, Mumtaz M, Adams DH, Deeb GM, Maini B, Gada H, et al; SURTAVI Investigators. Surgical or transcatheter aortic-valve replacement in intermediate-risk patients. *N Engl J Med*. 2017;376:1321–1331. doi: 10.1056/NEJMoa1700456
15. Mack MJ, Leon MB, Thourani VH, Makkar R, Kodali SK, Russo M, Kapadia SR, Malaisrie SC, Cohen DJ, Pibarot P, et al; PARTNER 3 Investigators. Transcatheter aortic-valve replacement with a balloon-expandable valve in low-risk patients. *N Engl J Med*. 2019;380:1695–1705. doi: 10.1056/NEJMoa1814052
16. Popma JJ, Deeb GM, Yakubov SJ, Mumtaz M, Gada H, O'Hair D, Bajwa T, Heiser JC, Merhi W, Kleiman NS, et al; Evolut Low Risk Trial Investigators. Transcatheter aortic-valve replacement with a self-expanding valve in low-risk patients. *N Engl J Med*. 2019;380:1706–1715. doi: 10.1056/NEJMoa1816885
17. Kim WK, Liebetrau C, Fischer-Rasokat U, Renker M, Rolf A, Doss M, Möllmann H, Nef H, Walther T, Hamm CW. Challenges of recognizing bicuspid aortic valve in elderly patients undergoing TAVR. *Int J Cardiovasc Imaging*. 2020;36:251–256. doi: 10.1007/s10554-019-01704-8
18. Makkar RR, Yoon SH, Chakravarty T, Kapadia SR, Krishnaswamy A, Shah PB, Kaneko T, Skipper ER, Rinaldi M, Babaliaros V, et al. Association between transcatheter aortic valve replacement for bicuspid vs tricuspid aortic stenosis and mortality or stroke among patients at low surgical risk. *JAMA*. 2021;326:1034–1044. doi: 10.1001/jama.2021.13346
19. Mylotte D, Lefevre T, Sondergaard L, Watanabe Y, Modine T, Dvir D, Bosmans J, Tchetche D, Kornowski R, Sinning J-M, et al. Transcatheter aortic valve replacement in bicuspid aortic valve disease. *J Am Coll Cardiol*. 2014;64:2330–2339. doi: 10.1016/j.jacc.2014.09.039
20. Yoon SH, Lefevre T, Ahn JM, Perlman GY, Dvir D, Latib A, Barbanti M, Deuschl F, De Backer O, Blanke P, et al. Transcatheter aortic valve replacement with early- and new-generation devices in bicuspid aortic valve stenosis. *J Am Coll Cardiol*. 2016;68:1195–1205. doi: 10.1016/j.jacc.2016.06.041
21. Makkar RR, Yoon SH, Leon MB, Chakravarty T, Rinaldi M, Shah PB, Skipper ER, Thourani VH, Babaliaros V, Cheng W, et al. Association between transcatheter aortic valve replacement for bicuspid vs tricuspid aortic stenosis and mortality or stroke. *JAMA*. 2019;321:2193–2202. doi: 10.1001/jama.2019.7108
22. Forrest JK, Kaple RK, Ramlawi B, Gleason TG, Meduri CU, Yakubov SJ, Jilaihawi H, Liu F, Reardon MJ. Transcatheter aortic valve replacement in bicuspid versus tricuspid aortic valves from the STS/ACC TVT Registry. *JACC Cardiovasc Interv*. 2020;13:1749–1759. doi: 10.1016/j.jcin.2020.03.022
23. Halim SA, Edwards FH, Dai D, Li Z, Mack MJ, Holmes DR, Tuzcu EM, Thourani VH, Harrison JK, Brennan JM. Outcomes of transcatheter aortic valve replacement in patients with bicuspid aortic valve disease. *Circulation*. 2020;141:1071–1079. doi: 10.1161/CIRCULATIONAHA.119.040333
24. Tarantini G, Fabris T. Transcatheter aortic valve replacement for bicuspid aortic valve stenosis. *Circ Cardiovasc Interv*. 2021;14:e009827. doi: 10.1161/CIRCINTERVENTIONS.120.009827
25. Xiong TY, Ali WB, Feng Y, Hayashida K, Jilaihawi H, Latib A, Lee MK, Leon MB, Makkar RR, Modine T, et al. Transcatheter aortic valve implantation in patients with bicuspid valve morphology: a roadmap towards standardization. *Nat Rev Cardiol*. 2023;20:52–67. doi: 10.1038/s41569-022-00734-5
26. Mangieri A, Tchetche D, Kim WK, Pagnesi M, Sinning J-M, Landes U, Kornowski R, De Backer O, Nickenig G, Ielasi A, et al. Balloon versus self-expandable valve for the treatment of bicuspid aortic valve stenosis. *Circ Cardiovasc Interv*. 2020;13:e008714. doi: 10.1161/CIRCINTERVENTIONS.119.008714
27. von Elm E, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP; STROBE Initiative. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *J Clin Epidemiol*. 2008;61:344–349. doi: 10.1016/j.jclinepi.2007.11.008
28. Genéréux P, Piazza N, Alu MC, Nazif T, Hahn RT, Pibarot P, Bax JJ, Leipsic JA, Blanke P, Blackstone EH, et al; VARC-3 Writing Committee. Valve Academic Research Consortium 3: updated endpoint definitions for aortic valve clinical research. *J Am Coll Cardiol*. 2021;77:2717–2746. doi: 10.1016/j.jacc.2021.02.038
29. Austin PC. Balance diagnostics for comparing the distribution of baseline covariates between treatment groups in propensity-score matched samples. *Stat Med*. 2009;28:3083–3107. doi: 10.1002/sim.3697
30. Harrell FE Jr. *Multivariable Modeling Strategies*. Springer; 2015.
31. Therneau TM, Grambsch PM. *Modeling Survival Data: Extending the Cox Model*. Springer-Verlag; 2000.
32. van Buuren S, Groothuis-Oudshoorn K. Multivariate imputation by chained equations in R. *J Stat Softw*. 2011;45:1–67. doi: 10.18637/jss.v045.i03
33. White IA, Royston P, Wood AM. Multiple imputations using chained equation. *Stat Med*. 2011;30:377–399. doi: 10.1002/sim.4067
34. van Buuren S. *Flexible Imputation of Missing Data*. Chapman and Hall/CRC; 2018.
35. Rosenbaum PR, Rubin DB. The central role of the propensity score in observational studies for causal effects. *Biometrika*. 1983;70:41–55. doi: 10.2307/2335942

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36. Rubin DB. Matching to remove bias in observational studies. *Biometrics*. 1973;29:159–184. doi: 10.2307/2529684
37. Austin PC. A comparison of 12 algorithms for matching on the propensity score. *Stat Med*. 2014;33:1057–1069. doi: 10.1002/sim.6004
38. Austin PC. An introduction to propensity score methods for reducing the effects of confounding in observational studies. *Multivariate Behav Res*. 2011;46:399–424. doi: 10.1080/00273171.2011.568786
39. Austin PC. The performance of different propensity score methods for estimating marginal hazard ratios. *Stat Med*. 2013;32:2837–2849. doi: 10.1002/sim.5705
40. Elze MC, Gregson J, Baber U, Williamson E, Sartori S, Mehran R, Nichols M, Stone GW, Pocock SJ. Comparison of propensity score methods and covariate adjustment. *J Am Coll Cardiol*. 2017;69:345–357. doi: 10.1016/j.jacc.2016.10.060
41. Granger E, Sergeant JC, Lunt M. Avoiding pitfalls when combining multiple imputation and propensity scores. *Stat Med*. 2019;38:5120–5132. doi: 10.1002/sim.8355
42. Bates D, Mächler M, Bolker BM, Walker SC. Fitting linear mixed-effects models using lme4. *J Stat Softw*. 2015;67:1–48. doi: 10.18637/jss.v067.i01
43. Austin PC. A tutorial on multilevel survival analysis: methods, models and applications. *Int Stat Rev*. 2017;85:185–203. doi: 10.1111/insr.12214
44. Therneau TM, Grambsch PM, Pankratz VS. Penalized survival models and frailty. *J Comput Graph Stat*. 2003;12:156–175. doi: 10.1198/1061860031365
45. Hirsch K, Wienke A. Software for semiparametric shared gamma and log-normal frailty models. *Comput Methods Programs Biomed*. 2012;107:582–597. doi: 10.1016/j.cmpb.2011.05.004
46. Wienke A. *Frailty Models in Survival Analysis*. Chapman & Hall/CRC; 2010.
47. Duchateau L, Janssen P. *The Frailty Model*. Springer-Verlag; 2008.
48. Funk MJ, Westreich D, Wiesen C, Sturmer T, Brookhart MA, Davidian M. Doubly robust estimation of causal effects. *Am J Epidemiol*. 2011;173:761–767. doi: 10.1093/aje/kwq439
49. Austin PC, Stuart EA. Moving towards best practice when using inverse probability of treatment weighting using the propensity score to estimate causal treatment effects in observational studies. *Stat Med*. 2015;34:3661–3679. doi: 10.1002/sim.6607
50. Michelena HI, Khanna AD, Mahoney D, Margaryan E, Topilsky Y, Suri RM, Eidem B, Edwards WD, Sundt TM, Enriquez-Sarano M. Incidence of aortic complications in patients with bicuspid aortic valves. *JAMA*. 2011;306:1104–1112. doi: 10.1001/jama.2011.1286
51. Michel JM, Frangieh AH, Giacoppo D, Alvarez-Covarrubias HA, Pellegrini C, Rheude T, Deutsch O, Mayr NP, Rumpf PM, Stähli BE, et al. Safety and efficacy of minimalist transcatheter aortic valve implantation using a new-generation balloon-expandable transcatheter heart valve in bicuspid and tricuspid aortic valves. *Clin Res Cardiol*. 2021;110:1993–2006. doi: 10.1007/s00392-021-01935-7
52. Montalto C, Sticchi A, Crimi G, Laricchia A, Khokhar AA, Giannini F, Reimers B, Colombo A, Latib A, Waksman R, et al. Outcomes after transcatheter aortic valve replacement in bicuspid versus tricuspid anatomy: a systematic review and meta-analysis. *JACC Cardiovasc Interv*. 2021;14:2144–2155. doi: 10.1016/j.jcin.2021.07.052
53. Ueshima D, Nai Fovino L, Brener SJ, Fabris T, Scotti A, Barioli A, Giacoppo D, Pavei A, Fraccaro C, Napodano M, et al. Transcatheter aortic valve replacement for bicuspid aortic valve stenosis with first- and new-generation bioprostheses: a systematic review and meta-analysis. *Int J Cardiol*. 2020;298:76–82. doi: 10.1016/j.ijcard.2019.09.003
54. Forrest JK, Ramlawi B, Deeb GM, Zahr F, Song HK, Kleiman NS, Chetcuti SJ, Michelena HI, Mangi AA, Skiles JA, et al. Transcatheter aortic valve replacement in low-risk patients with bicuspid aortic valve stenosis. *JAMA Cardiol*. 2021;6:50–57. doi: 10.1001/jamacardio.2020.4738
55. Deeb GM, Reardon MJ, Ramlawi B, Yakubov SJ, Chetcuti SJ, Kleiman NS, Mangi AA, Zahr F, Song HK, Gada H, et al. Propensity-matched 1-year outcomes following transcatheter aortic valve replacement in low-risk bicuspid and tricuspid patients. *JACC Cardiovasc Interv*. 2022;15:511–522. doi: 10.1016/j.jcin.2021.10.027
56. Yoon SH, Kim WK, Dhoble A, Milhorini Pio S, Babaliaros V, Jilaihawi H, Pilgrim T, De Backer O, Bleiziffer S, Vincent F, et al; Bicuspid Aortic Valve Stenosis Transcatheter Aortic Valve Replacement Registry Investigators. Bicuspid aortic valve morphology and outcomes after transcatheter aortic valve replacement. *J Am Coll Cardiol*. 2020;76:1018–1030. doi: 10.1016/j.jacc.2020.07.005
57. Faroux L, Chen S, Muntane-Carol G, Regueiro A, Philippon F, Sondergaard L, Jørgensen TH, Lopez-Aguilera J, Kodali S, Leon M, et al. Clinical impact of conduction disturbances in transcatheter aortic valve replacement recipients: a systematic review and meta-analysis. *Eur Heart J*. 2020;41:2771–2781. doi: 10.1093/eurheartj/ehz924
58. Khan A, Ramsey K, Ballard C, Armstrong E, Burchill LJ, Menashe V, Pantely G, Broberg CS. Limited Accuracy of Administrative Data for the Identification and Classification of Adult Congenital Heart Disease. *J Am Heart Assoc*. 2018;7:e007378. doi: 10.1161/jaha.117.007378
59. Khan MS, Usman MS, Van Spall HGC, Greene SJ, Baqal O, Felker GM, Bhatt DL, Januzzi JL, Butler J. Endpoint adjudication in cardiovascular clinical trials. *Eur Heart J*. 2023;44:4835–4846. doi: 10.1093/eurheartj/ehad718
60. Sievers HH, Schmidtke C. A classification system for the bicuspid aortic valve from 304 surgical specimens. *J Thorac Cardiovasc Surg*. 2007;133:1226–1233. doi: 10.1016/j.jtcvs.2007.01.039
61. Makkar RR, Cheng W, Waksman R, Satler LF, Chakravarty T, Groh M, Abernethy W, Russo MJ, Heimansohn D, Hermiller J, et al. Self-expanding intra-annular versus commercially available transcatheter heart valves in high and extreme risk patients with severe aortic stenosis (PORTICO IDE): a randomised, controlled, non-inferiority trial. *Lancet*. 2020;396:669–683. doi: 10.1016/S0140-6736(20)31358-1
62. Lanz J, Kim WK, Walther T, Burgdorf C, Möllmann H, Linke A, Redwood S, Thilo C, Hilker M, Joner M, et al; SCOPE I Investigators. Safety and efficacy of a self-expanding versus a balloon-expandable bioprosthesis for transcatheter aortic valve replacement in patients with symptomatic severe aortic stenosis: a randomised non-inferiority trial. *Lancet*. 2019;394:1619–1628. doi: 10.1016/S0140-6736(19)32220-2
63. Tamburino C, Bleiziffer S, Thiele H, Scholtz S, Hildick-Smith D, Cunnington M, Wolf A, Barbanti M, Tchetchè D, Garot P, et al. Comparison of self-expanding bioprostheses for transcatheter aortic valve replacement in patients with symptomatic severe aortic stenosis: SCOPE 2 randomized clinical trial. *Circulation*. 2020;142:2431–2442. doi: 10.1161/CIRCULATIONAHA.120.051547
64. Denz R, Klaassen-Mielke R, Timmesfeld N. A comparison of different methods to adjust survival curves for confounders. *Stat Med*. 2023;42:1461–1479. doi: 10.1002/sim.9681
65. Morisot A, Bessaoud F, Landais P, Rebillard X, Tretarre B, Daures JP. Prostate cancer: net survival and cause-specific survival rates after multiple imputation. *BMC Med Res Methodol*. 2015;15:54. doi: 10.1186/s12874-015-0048-4
66. Marshall A, Altman DG, Holder RL, Royston P. Combining estimates of interest in prognostic modelling studies after multiple imputation: current practice and guidelines. *BMC Med Res Methodol*. 2009;9:57. doi: 10.1186/1471-2288-9-57
67. Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J R Stat Soc Ser B Stat Methodol*. 1995;57:289–300. doi: 10.1111/j.2517-6161.1995.tb02031.x

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